

Effectiveness of Dader Method for Pharmaceutical Care on Control of Blood Pressure and Total Cholesterol in Outpatients with Cardiovascular Disease or Cardiovascular Risk: EMDADER-CV Randomized Controlled Trial

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ABSTRACT

BACKGROUND: Although some studies have demonstrated that pharmacist intervention can improve drug therapy among patients with cardiovascular disease (CVD), more evidence derived from randomized controlled trials (RCTs) is needed, including assessment of the effect of community pharmacist interventions in patients with CVD.

OBJECTIVE: To assess the effectiveness of the Dader Method for pharmaceutical care on achieving therapeutic goals for blood pressure (BP), total cholesterol (TC), and both BP and TC (BP/TC) in patients with CVD and/or high or intermediate cardiovascular (CV) risk attending community pharmacies in Spain.

METHODS: Patients aged 25 to 74 years attending community pharmacies with a prescription for at least 1 drug indicated for CVD or CV risk factors were randomized to 2 groups: an intervention group that received pharmaceutical care, which was provided by specially trained pharmacists working in collaboration with physicians, and a control group that received usual care (routine dispensing counseling) and verbal and written counseling regarding CVD prevention. Patients were recruited from December 2005 to September 2006, and both groups were followed for 8 months. Study outcomes were assessed at baseline and at 16 and 32 weeks after randomization. The primary outcome measures were the proportions of patients achieving BP, TC, and BP/TC therapeutic goals (BP lower than 140/90 mm Hg for patients with uncomplicated hypertension and lower than 130/80 mm Hg for patients with diabetes, chronic kidney disease, or history of myocardial infarction or stroke; TC lower than 200 mg per dL for patients without CVD and lower than 175 mg per dL for patients with CVD). Secondary outcomes were mean BP and TC values. BP was assessed manually by the pharmacist after a 10-minute rest in the supine position. This measurement was performed twice for every participant, and the average of the 2 measurements was calculated. TC was measured by the pharmacist during the study visit using the enzymatic dry method. Statistical analyses were performed using 2-tailed McNemar tests, Pearson chi-square tests, and Student's t-tests; $P < 0.05$ was considered statistically significant.

RESULTS: 714 patients were included in the study (356 intervention, 358 control), and the mean [SD] age was 62.8 [8.1] years. The 2 groups were similar at baseline in clinical and demographic characteristics, including the proportion of patients at therapeutic goals for BP, TC, and BP/TC. After 8 months of follow-up, there were statistically significant differences in favor of pharmaceutical care in the proportions of patients who achieved therapeutic goals for BP (52.5% vs. 43.0%, $P = 0.017$), TC (56.5% vs. 44.1%, $P = 0.001$), and BP/TC (37.1% vs. 21.8%, $P < 0.001$).

CONCLUSION: Compared with usual care plus written education, pharmaceutical care focused on patient evaluation and follow-up in collaboration with physicians improved the achievement of BP, TC, and BP/TC treatment goals in patients with CVD and/or high or intermediate CV risk attending community pharmacies in Spain.

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What is already known about this subject

- Despite the availability of evidence-based guidelines for treatment and prevention of cardiovascular disease (CVD), many patients do not achieve therapeutic goals. In studies by Banegas et al. (1998) and Rodríguez-Roca et al. (2005), only 15.5%-33.5% of patients receiving drug therapy for hypertension achieved their blood pressure (BP) treatment goals. Similarly, Olson et al. (2001) reported that only 21% (range 18%-35%) of patients receiving lipid-lowering therapy achieved their cholesterol targets.
- Community pharmacists working in cooperation with patients and other health care professionals could contribute to achieving desired outcomes in patients with chronic diseases. A prospective randomized controlled trial (RCT) by Morgado et al. (2011), conducted in a secondary care hypertension/dyslipidemia outpatient clinic, found BP control rates of 63.3% (62/98) in patients receiving pharmaceutical care, compared with 43.4% (40/99) in the usual care group. An RCT of patients at high cardiovascular (CV) risk by Tsuyuki et al. (2002) found the primary endpoint (performance of a fasting cholesterol panel by the physician or addition or increase in dose of cholesterol-lowering medication) was reached in 196 patients (57%) in a community pharmacy intervention group, compared with 102 (31%) in usual care. However, in a recent systematic review of studies evaluating community pharmacist interventions for preventing or managing diabetes, CVD, and/or major CV risk factors, Evans et al. (2011) concluded that there is a need for more high-quality studies of these interventions.

What is already known about this subject (continued)

- The Dader Method for pharmaceutical care includes patient education about CV drugs, completion of a drug therapy profile and/or drug history, assessment of drug compliance, patient counseling about lifestyle modifications, pharmacist-performed interventions not related to changes in drug therapy, and pharmacist-delivered treatment recommendations to physicians.
- To our knowledge, there is a lack of large RCTs designed to evaluate the effect of community pharmacist interventions on achieving therapeutic goals for BP, total cholesterol (TC), and both BP and TC (BP/TC) in patients with CVD or CV risk factors.

What this study adds

- The EMDADER-CV (Efecto del Método Dáder de Seguimiento Farmacoterapéutico en el riesgo cardiovascular de pacientes con factores de riesgo o enfermedad cardiovascular [Effectiveness of Dader Method for Pharmaceutical Care on Control of Blood Pressure and Total Cholesterol in Outpatients with Cardiovascular Disease or Cardiovascular Risk]) study was a large RCT designed to assess the effect of the Dader Method for pharmaceutical care on the achievement of therapeutic goals for BP, TC, and BP/TC in community pharmacy patients (n=714) with CVD or CV risk factors.
- After 8 months follow-up, there were statistically significant differences in favor of the intervention group (n=356) compared with the control group (n=358) in the percentage of patients who achieved therapeutic goals for systolic BP (54.5% vs. 45.5%, odds ratio [OR]=1.43, 95% CI=1.06-1.94); BP (52.5% vs. 43.0%, OR=1.47, 95% CI=1.08-1.99); TC (56.5% vs. 44.1%, OR=1.64, 95% CI=1.21-2.23); and BP/TC (37.1% vs. 21.8%, OR=2.12, 95% CI=1.50-2.98).

Cardiovascular disease (CVD) is the leading cause of mortality and disability worldwide.¹ It is considered an important public health problem because of its high prevalence, mortality, and negative effects on health and quality of life.² Therefore, prevention and treatment of CVD are essential elements of chronic disease control.³ Despite recent advances, evidence of benefits obtained with treatment and prevention of CVD, and the availability of treatment guidelines supported by the results of clinical studies,⁴ there is a gap between evidence-based recommendations and routine clinical practice.⁵ As a consequence, many patients do not receive necessary therapeutic interventions or do not achieve their therapeutic goals.⁶ For instance, in studies by Banegas et al. (1998) and Rodríguez-Roca et al. (2005), only 15.5%-33.5% of patients receiving drug therapy for hypertension achieved their blood pressure (BP) treatment goals,^{7,8} rates similar to those of patients with hypercholesterolemia.^{6,9} Olson et al. (2001) reported that only 21% (range 18% to 35%) of patients receiving lipid-lowering therapy achieved their cholesterol targets.⁶

In Spain, it is estimated that only 10% of patients with high or very high cardiovascular (CV) risk have major CV risk factors, such as hypertension, hypercholesterolemia, diabetes, and smoking, under control.¹⁰ Similarly, an analysis of National Health and Nutrition Examination Survey (NHANES) data that assessed the prevalence, treatment, and control of combined hypertension and hypercholesterolemia in the United States among a sample of 2,864 adults aged 20 years or older found that, among 638 adults, the percentage controlled for combined hypertension and hypercholesterolemia was only 9%.¹¹ Accordingly, there is a need for exploring different strategies to increase the percentage of patients achieving therapeutic goals for CV risk factors.

Community pharmacists, in cooperation with patients and other clinicians, could monitor therapeutic plans and contribute to the achievement of clinical outcomes in patients with chronic diseases.¹² Some studies have shown that pharmacist participation in designing, implementing, and monitoring therapeutic programs in patients with CVD or CV risk factors could (a) improve the patient's knowledge,¹³ (b) facilitate adoption of and outcomes from lifestyle interventions,¹⁴ (c) improve the identification of candidates and the results of primary¹⁵ or secondary¹⁶ prevention, and (d) increase the percentage of patients with high CV risk who achieve the goals related to BP¹⁷⁻²⁰ and lipid^{9,21} levels. In contrast, other randomized controlled trials (RCTs) have found that community pharmacist intervention did not lead to improvements in primary outcome measures (e.g., hospital admissions²² or the proportion of patients receiving appropriate medication²³).

In a systematic review and assessment of the quality of studies evaluating community pharmacist interventions in preventing or managing diabetes or CVD and/or major CV risk factors, Evans et al. (2011) found that study quality was generally poor (poor design and/or poor reporting), most studies assessed only biomarkers, and none of the studies demonstrated any benefits in major health outcomes (e.g., hospital admissions or mortality). They concluded that there is a need for more high-quality studies of community pharmacist interventions in patients with CVD.²⁴

The objective of the present study was to assess the effectiveness of the Dader Method for pharmaceutical care on the achievement of therapeutic goals for BP, total cholesterol (TC), and both BP and TC (BP/TC) in patients with CVD and/or high or intermediate CV risk attending community pharmacies in Spain. TC was chosen as a therapeutic goal measure instead of low-density lipoprotein cholesterol (LDL-C) because TC is measurable during community pharmacy visits, whereas LDL-C is not.

Methods

Detailed methods of EMDADER-CV (Efecto del Método Dáder de Seguimiento Farmacoterapéutico en el riesgo cardiovascular de pacientes con factores de riesgo o enfermedad

cardiovascular [Effectiveness of Dader Method for Pharmaceutical Care on Control of Blood Pressure and Total Cholesterol in Outpatients with Cardiovascular Disease or Cardiovascular Risk]) have been published previously (in Spanish only).²⁵ Overall, the EMDADER-CV was an RCT designed to compare the effectiveness of the Dader Method for pharmaceutical care with that of the usual care process in a community pharmacy setting in Spain. Patients aged 25 to 74 years and attending a community pharmacy with a prescription for at least 1 drug indicated for CVD or CV risk factors were randomized to 1 of 2 groups: an intervention group that received pharmaceutical care, which was provided by specially trained pharmacists, or a control group that received usual care (routine dispensing and oral counseling regarding drugs) and verbal and written counseling regarding CV disease prevention (information about CVD and CV risk factors). Prior to all study visits, described in detail below, patients in both groups received telephone calls from the study pharmacist, who encouraged their attendance at the visits.

Community Pharmacist Recruitment

Sixty community pharmacists from 13 provinces of Spain were invited to participate in the study: 45 were taking the certificate program in pharmaceutical care; 10 had taken the certificate program in pharmaceutical care; and 5 had a master's degree in pharmaceutical care from the University of Granada. Of these, all pharmacists who agreed to attend an informational meeting and to participate in the study were provided with an 8-hour training course on pharmacist interventions in CVD. This course included lectures on CVD, CV risk factors, CVD prevention, and intervention and monitoring of patients with risk factors or CVD, focusing on therapeutic goals for BP and TC, according to each patient's clinical condition.

Patient Recruitment and Group Assignment

Researchers recruited patients who met the following inclusion criteria: (a) aged 25 to 74 years; (b) presented at the pharmacy with a prescription for at least 1 drug indicated for hypertension, hypercholesterolemia, CVD prophylaxis, or type 2 diabetes; and (c) had a high or moderate CV risk according to the Systematic Coronary Risk Evaluation (SCORE) system and/or the Wilson-Grundy method, adapted for Spain.²⁶ Participants were excluded from the study if they had a BP of 180/110 millimeters mercury (mm Hg) or higher; history of myocardial infarction in the previous 3 months; a terminal disease (e.g., terminal cancer or chronic kidney disease on dialysis); an intellectual or physical disability that prevented them from participating in the study; or if they were currently included in a cardiac rehabilitation program. Patients were randomized to 1 of 2 groups, intervention or control, and were followed for 8 months. An 8-month time frame was chosen because it permitted capture of at least 3 stable changes in TC and BP values associated with drug therapy and lifestyle modifications.

Randomization

Patients were assigned to the intervention or the control group using a computer-generated randomization schedule that was designed using the Linux operating system, the compiler GCC version 3.3.5 (Debian 1:3.3.5-12; free software available at www.debian.org), and the programming language C++. The program generated a sequence of 1,000 random numbers with no repetition (1 to 1,000) and assigned the code ONE to 500 numbers and the code ZERO to the other 500. Finally, it grouped the previous 1,000 random numbers (coded ONE or ZERO) into 50 groups of 20. Each pharmacy entered into the study when the pharmacist submitted by fax or email the record of the first patient who fulfilled the inclusion criteria. Once the study's coordinator verified the fulfillment of the inclusion criteria, he randomly assigned 1 of the mentioned 50 groups to the pharmacy, providing it with a sequence of 20 codes (ONE or ZERO) that determined which patient was assigned to the intervention group or the control group.

Intervention Group: The Dader Method for Pharmaceutical Care

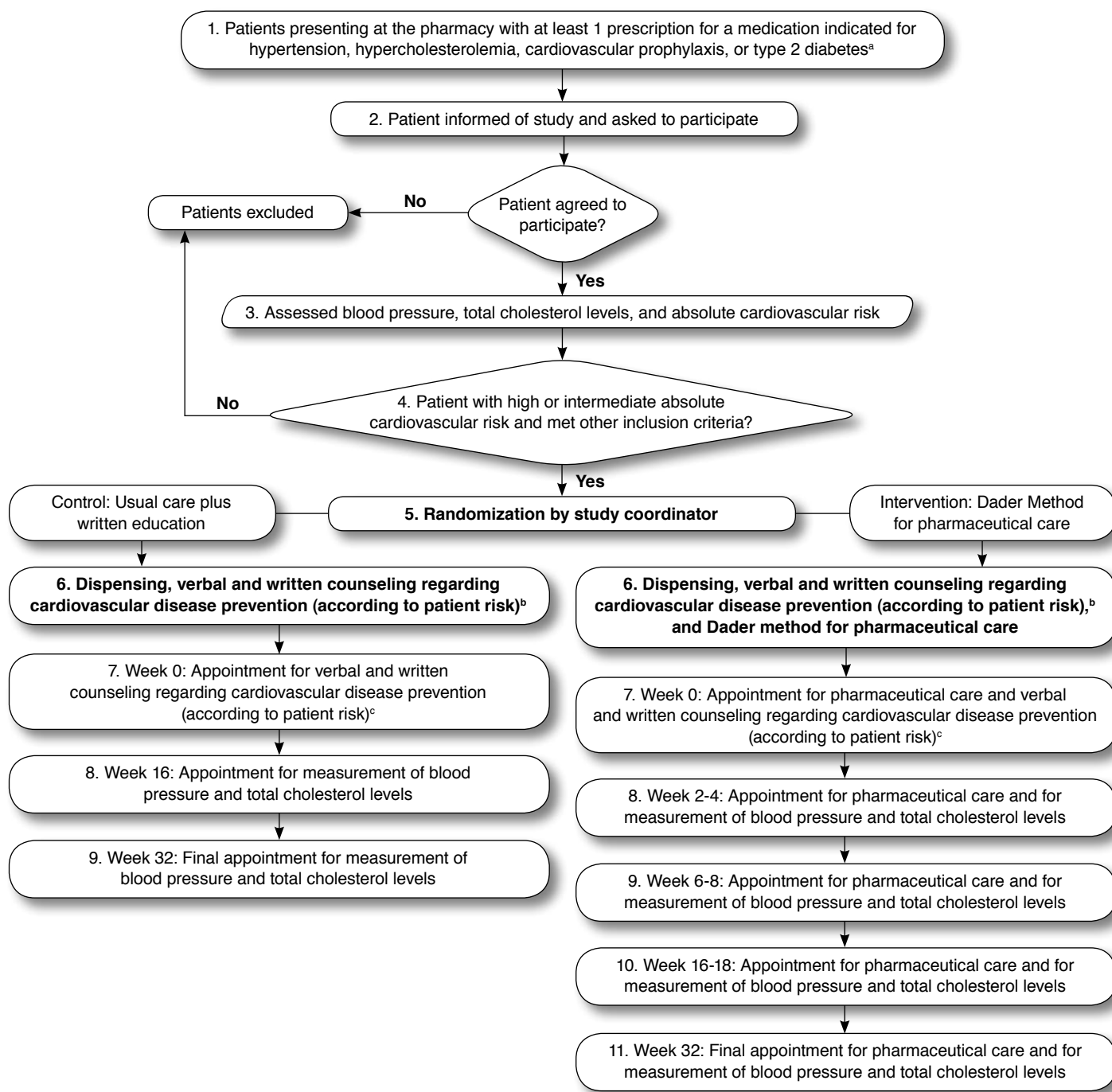
The Dader Method for pharmaceutical care is a systematic process developed by the Research Group of Pharmaceutical Care at the University of Granada, Spain.²⁷ The intervention is based on the use of pharmacotherapy records, evaluation of an assessment form that includes all CV health problems and the CV drugs used to treat these medical problems, and their assessment on a specific date. This assessment is used to identify (a) any potential or actual patient health outcomes that are not consistent with the objectives of pharmacotherapy and are associated with the use of medicines (negative outcomes associated with medication [NOM]) and (b) situations in which the use of medicines caused or may cause the appearance of an NOM (drug-related problems [DRP]).²⁸ Once the relevant problems are identified, the necessary interventions to patients or to physicians are carried out to solve the identified NOM and are followed by a subsequent assessment of the achieved outcomes.

In the present study, pharmaceutical care was provided on a regular basis according to a systematic and documented method and carried out in collaboration with patients and physicians (Figure 1).²⁸ Thus, in the intervention group, pharmacists did the following:²⁷

- (a) Obtained patient data related to CV medical problems and current drug therapy. Patient-specific data were obtained by interviewing the patient and reviewing the drug and clinical records (mainly drug history and results of clinical laboratory tests that patients presented during the interview).
- (b) Used the collected data to complete the assessment form, which was interpreted and evaluated once all the necessary information was added. The key element to completing the assessment form was the pairing of the health problems

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FIGURE 1 General Overview of EMDADER-CV



^aExamples for hypertension included enalapril, amlodipine, valsartan, hydrochlorothiazide, bisoprolol, etc.; for hypercholesterolemia: atorvastatin, lovastatin, pravastatin, gemfibrozil, etc.; for cardiovascular prophylaxis: aspirin 100 mg, clopidogrel, etc.; for type 2 diabetes: glibenclamide, gliclazide, metformin, pioglitazone, rosiglitazone, etc.

^bA semivalidated cardiovascular disease knowledge questionnaire was applied to all patients in the visits that took place during weeks 0, 16, and 32. During the same visits, in 85 patients (41 control group and 44 intervention group), adherence to treatment was assessed using the Morisky-Green-Levine test.

^cDuring week 0, randomization took place.

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with the current drug therapy, which provided a global view of the patient's health status and its relationship with the drugs used.

- (c) Evaluated the patient's drug therapy outcomes. The aim of this stage was to assess whether the desired treatment goals for BP and TC were achieved. For patients whose goals were not yet achieved, the pharmacist developed therapeutic plans that included interventions with the aim of achieving the desired clinical outcome.
- (d) Conducted an intervention intended to directly prevent or resolve an NOM.²⁹ Once the pharmacist identified concerns about the medical problems and current drug therapy, he or she interpreted and analyzed this information in the context of the clinical condition illustrated by the assessment form, taking into account factors such as the patient's CV risk, the type of CV prevention, and the magnitude of increase in BP or in TC. If the aim of the intervention was to modify a problem with lifestyle or with use of a medication, the recipient of the intervention was the patient. If the aim of the intervention was to modify drug therapy in either a quantitative way (e.g., modifying dosage or frequency) or qualitative way (e.g., adding or changing any drug), the recipient of the intervention was the physician.
- (e) Completed a new assessment form. Completion of an intervention should have generated a change in the patient's assessment. Depending on whether an NOM still existed, the therapeutic plan was carried out. Examples included (i) informing the physician that drug therapy for hypertension or for hypercholesterolemia was not effective and that the patient therefore may need a modification in drug therapy and (ii) providing the physician with information related to a need for drug therapy to address medical problems or for CVD prophylaxis.

Patients assigned to the intervention group were provided with verbal and written counseling regarding CVD prevention (according to patient risk). To assess the results of interventions on BP and/or TC, patients had at least 5 appointments with the pharmacist on weeks 0 (the week during which randomization took place), 4-6, 8-10, 14-16, and 32. BP, TC, body mass index (BMI), smoking status, and exercise routines were evaluated during those appointments. The timing of the intermediate appointments (weeks 4-6, 8-10, 14-16) was flexible, depending on the amount of time necessary to assess the results of interventions that had been performed.

Control Group: Usual Care (Routine Dispensing), Verbal and Written Education

Due to the nature of the intervention, participant blinding was not possible. There was no "placebo" treatment, and after randomization, patients were informed of their group assignments. The control group received the usual care provided by the pharmacist (routine dispensing, including oral counseling

regarding drugs) and written information and education on CV risk factors and CVD (Figure 1). The written material was a brochure designed for this purpose,³⁰ which focused on the goals and importance of adherence with pharmacological and nonpharmacological interventions to achieve treatment goals. In addition to a visit during week 0 (the week during which randomization took place), patients met again with the pharmacist on weeks 16 and 32. At each appointment, BP, TC, BMI, smoking status, and exercise routines were assessed.

In 85 patients (41 from the control group and 44 from the intervention group) adherence to treatment was assessed using the Morisky-Green-Levine test on weeks 0, 16, and 32. In addition, a semivalidated CVD knowledge questionnaire was applied to all patients at weeks 0, 16, and 32.

Sample Size Calculation

Based on the studies by Olson et al.,⁶ Banegas et al.,⁷ and Rodríguez-Roca et al.,⁸ we assumed that in the control group only 30% of patients would achieve the desired values of BP and TC and that in the intervention group this percentage would be 40%. With that difference in mind, a significance level of 0.05 (2-tailed) and a power of 80%, we estimated a required sample size of 752 patients (376 in each group). To increase statistical power for secondary outcomes, we tried to achieve a sample size of 1,000 patients (500 in each group), 20 patients from 50 pharmacies.

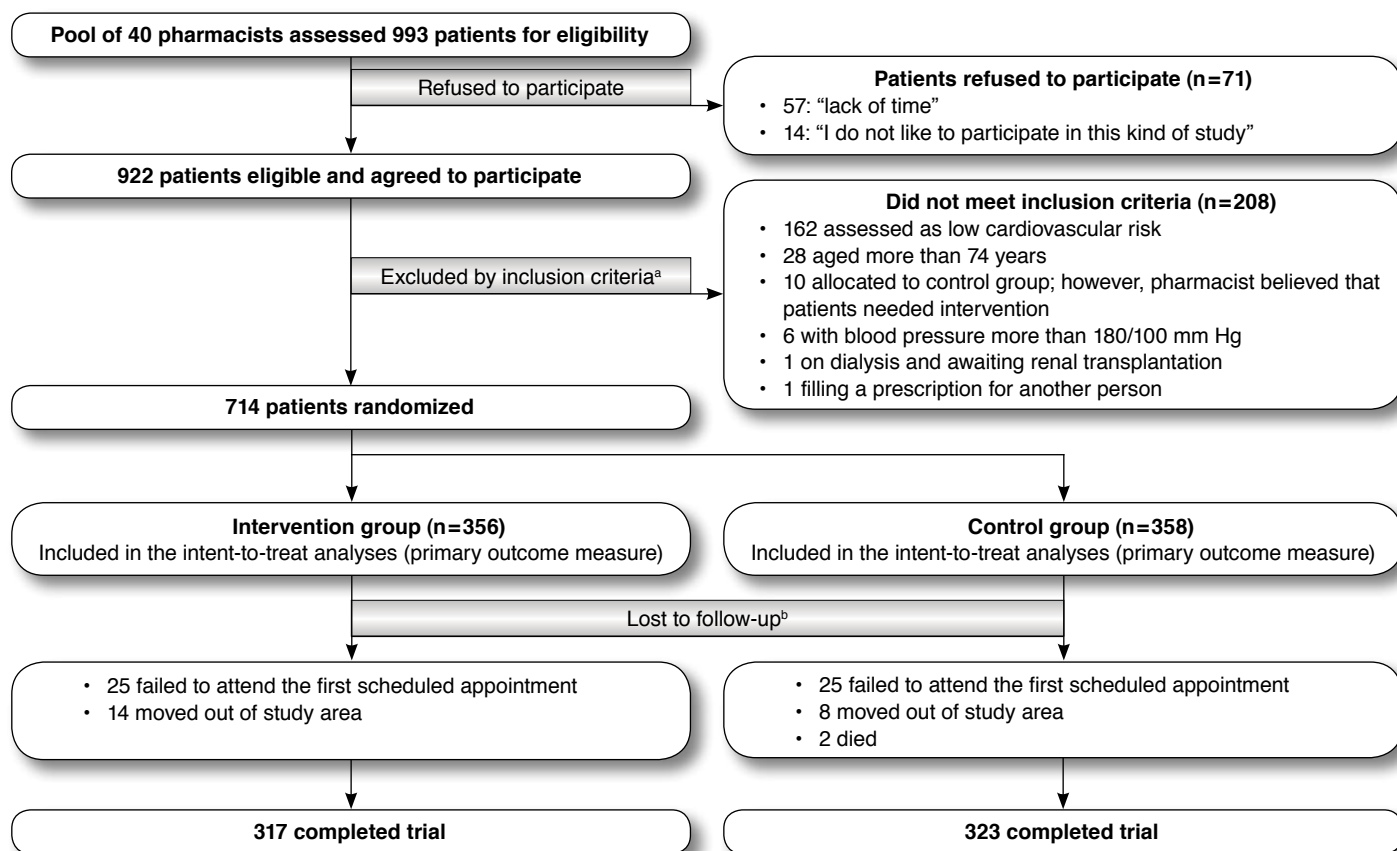
Outcome Data and Analysis

The primary outcomes were the proportions of patients achieving BP, TC, and BP/TC therapeutic goals, according to their clinical conditions: BP lower than 140/90 mm Hg for patients with uncomplicated hypertension; lower than 130/80 mm Hg for patients with diabetes, chronic kidney disease, history of myocardial infarction, or history of stroke; and total cholesterol lower than 200 milligrams per deciliter (mg per dL) for patients without CVD and lower than 175 mg per dL for patients with CVD.⁴ Secondary outcomes were mean BP and TC values.

BP was assessed manually by the pharmacist after a 10-minute rest in a supine position. This measurement was performed twice for every participant at each visit, and the average of the 2 measurements was calculated. TC was measured by the pharmacist during the visit using the enzymatic dry method. For this purpose, the pharmacist used the Reflotron system (Roche Diagnostic, SL, Barcelona, Spain), standardized according to the manufacturer's instructions. At baseline and endpoint, control and intervention groups were compared in terms of the proportion of patients at therapeutic goal for BP, TC, and BP/TC.

Statistical Analysis. Statistical analyses were performed using SPSS version 13.0 (IBM SPSS, Armonk, New York). Data were reported as means and standard deviations [SD] or as percentages. The Pearson chi-square test (between study

FIGURE 2 Flow of Patients Through the EMDADER-CV Study



^aNo exclusions were made for patients in a cardiac rehabilitation program, with intellectual or physical disability, or with history of myocardial infarction in the previous 3 months.

^bAnalyses assumed that all patients lost to follow-up from both the control and intervention groups did not achieve therapeutic goals for blood pressure or total cholesterol. EMDADER-CV = Efecto del Método Dáder de Seguimiento Farmacoterapéutico en el riesgo cardiovascular de pacientes con factores de riesgo o enfermedad cardiovascular (Effectiveness of Dader Method for Pharmaceutical Care on Control of Blood Pressure and Total Cholesterol in Outpatients with Cardiovascular Disease or Cardiovascular Risk); mm Hg = millimeters mercury.

groups) and the McNemar test (within-group changes from baseline to follow-up at 8 months) were used to compare proportions. The paired Student's t-test (within-group changes from baseline to follow-up at 8 months) or the independent sample Student's t-test (between study groups) were used to compare means; odds ratios (ORs) and 95% confidence intervals (CIs) were also estimated. Comparisons were analyzed using 2-tailed tests, and $P < 0.05$ was considered statistically significant.

Ethical Approval

Ethical approval for the study protocol was granted by the University of Granada's Human Research Ethical Committee, and voluntary written informed consent was required from all participants.

Results

Of the 60 community pharmacists (from 60 pharmacies) invited to participate in the study, only 40 (66.7%) participated in EMDADER-CV. The first patient was randomized in December 2005, and randomization was completed in September 2006. Follow-up was completed between August 2006 and June 2007. The 40 participating pharmacists assessed 993 patients for eligibility, but 71 patients refused to participate (Figure 2). From the remaining 922 patients, 208 did not meet the inclusion criteria, and 714 subjects were included and randomized: 356 were assigned to the intervention group and 358 to the control group. The mean [SD] age of the patients was 62.8 [8.1] years, 52.2% (373) were males, and 78.6% (561) had high CV risk (Table 1). Overall, the 2 groups were similar in age, male-female ratio, comorbidities, and other clinical conditions (all P values > 0.05 ; Table 1).

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TABLE 1 Demographic and Clinical Characteristics Related to Primary Goal at Baseline

Demographic and Clinical Variables	Total (N = 714)		Intervention Group (n = 356)		Control Group (n = 358)		P Value ^a
Age, mean [SD]	62.8	[8.1]	63.0	[8.3]	62.6	[8.0]	0.506
	%	(n)	%	(n)	%	(n)	
Gender							
Male	52.2	(373)	50.6	(180)	53.9	(193)	0.370
Female	47.8	(341)	49.4	(176)	46.1	(165)	
Education							
Elementary school or less	70.9	(506)	71.4	(254)	70.4	(252)	0.767
High school	15.8	(113)	14.9	(53)	16.8	(60)	
University education	13.3	(95)	13.8	(49)	12.8	(46)	
Marital status							
Married	76.6	(547)	73.6	(262)	79.6	(285)	0.058
Single	23.4	(167)	26.4	(94)	20.4	(73)	
Self-reported health status							
Fair, poor, or very poor	63.6	(454)	65.2	(232)	62.0	(222)	0.381
Excellent or good	36.4	(260)	34.8	(124)	38.0	(136)	
Current smoker	20.3	(145)	19.1	(68)	21.5	(77)	0.424
Current physical activity	52.9	(378)	54.8	(195)	51.1	(183)	0.328
Body mass index (kg/m²)							
Normal	14.1	(101)	14.1	(47)	16.1	(54)	0.665
Overweight	37.4	(267)	38.8	(138)	36.0	(129)	
Obesity	40.8	(291)	41.9	(149)	42.5	(152)	
Mean [SD] ^b	30.0	[5.0]	29.9	[4.9]	30.1	[5.1]	0.589
Comorbid and other clinical conditions							
	%	(n)	%	(n)	%	(n)	
Secondary prevention	28.6	(204)	29.8	(106)	27.4	(98)	0.478
• Myocardial infarction	11.2	(80)	11.8	(42)	10.6	(38)	0.616
• Ischemic heart disease	9.4	(67)	9.6	(34)	9.2	(33)	0.879
• Cerebrovascular disease	4.2	(30)	4.5	(16)	3.9	(14)	0.697
• Peripheral arterial disease	3.8	(27)	3.9	(14)	3.6	(13)	0.833
Diabetes	41.9	(299)	39.9	(142)	43.9	(157)	0.283
Heart failure	9.1	(65)	9.0	(32)	9.2	(33)	0.915
Hypertension	86.7	(619)	88.8	(316)	84.6	(303)	0.104
Dyslipidemia	71.3	(509)	71.3	(254)	71.2	(255)	0.972
Family dyslipidemia	1.7	(12)	1.7	(6)	1.7	(6)	0.992
Clinical condition that indicates high cardiovascular risk	64.1	(458)	64.3	(229)	64.0	(229)	0.920
High cardiovascular risk	78.6	(561)	77.8	(277)	79.3	(284)	0.621
Medium cardiovascular risk	21.4	(153)	22.2	(79)	20.7	(74)	

^aTests of between-group differences for control versus intervention groups: Pearson chi-square test to compare proportions and the Student's *t*-test to compare means.

^bn = 335 intervention, n = 334 control.

kg = kilograms; m² = squared meters; SD = standard deviation.

Primary Outcomes

At baseline in the intervention group, 129 (36.2%), 236 (66.3%), 117 (32.9%), 151 (42.4%), and 49 (13.8%) patients were at their therapeutic goals for systolic blood pressure (SBP), diastolic blood pressure (DBP), BP, TC, and BP/TC, respectively (Table 2). These percentages were not significantly different from the values in the control group, where 137 (38.3%), 242 (67.6%), 128 (35.8%), 140 (39.1%), and 51 (14.2%) patients were at their therapeutic goals for SBP (*P* = 0.575), DBP (*P* = 0.711), BP (*P* = 0.417), TC (*P* = 0.369), and BP/TC (*P* = 0.853), respectively. The distribution of patients who achieved their

therapeutic goals for BP, TC, and BP/TC was similar in both groups in baseline demographics, comorbidities, and other clinical conditions.

After 8 months of follow-up, the intervention group showed statistically significant increases in the percentages of patients who achieved therapeutic goals in SBP (18.3 percentage points, 95% CI = 12.2-24.4); BP (19.6 percentage points, 95% CI = 13.6-25.8); TC (14.1 percentage points, 95% CI = 8.4-19.7); and BP/TC (23.3 percentage points, 95% CI = 16.7-28.9; Table 2). However, the increase in the proportion of patients who achieved DBP therapeutic goals was not statistically significant (5.3 percentage points, 95% CI = -0.6-11.2).

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TABLE 2 Changes in Primary and Secondary Outcomes

Primary Outcomes	Baseline %	(n)	8-Month Follow-Up %	(n)	Difference (95% CI)	P Value ^a
Intervention group (n = 356 patients)						
Systolic blood pressure control	36.2	(129)	54.5	(194)	18.3 (12.2-24.4)	<0.001
Diastolic blood pressure control	66.3	(236)	71.6	(255)	5.3 (−0.6-11.2)	0.076
Blood pressure control ^b	32.9	(117)	52.5	(187)	19.6 (13.6-25.8)	<0.001
Total cholesterol control ^c	42.4	(151)	56.5	(201)	14.1 (8.4-19.7)	<0.001
Blood pressure and total cholesterol control ^{b,c}	13.8	(49)	37.1	(132)	23.3 (16.7-28.9)	<0.001
Control group (n = 358 patients)						
Systolic blood pressure control	38.3	(137)	45.5	(163)	7.2 (1.6-12.9)	0.012
Diastolic blood pressure control	67.6	(242)	70.4	(252)	2.8 (−2.5-8.1)	0.303
Blood pressure control ^b	35.8	(128)	43.0	(154)	7.2 (1.7-12.8)	0.011
Total cholesterol control ^c	39.1	(140)	44.1	(158)	5.0 (−0.2-10.3)	0.060
Blood pressure and total cholesterol control ^{b,c}	14.2	(51)	21.8	(78)	7.6 (3.1-12.0)	0.001
Between-group comparison at 8-month follow-up						
	Intervention Group %	(n)	Control Group %	(n)	Difference (95% CI)	P Value^a
Systolic blood pressure control	54.5	(194)	45.5	(163)	9.0 (1.6-16.3)	0.017
Diastolic blood pressure control	71.6	(255)	70.4	(252)	1.2 (−5.4-7.9)	0.716
Blood pressure control ^b	52.5	(187)	43.0	(154)	9.5 (2.2-16.8)	0.011
Total cholesterol control ^c	56.5	(201)	44.1	(158)	12.3 (5.0-19.6)	0.001
Blood pressure and total cholesterol control ^{b,c}	37.1	(132)	21.8	(78)	15.3 (8.7-21.9)	<0.001
Secondary Outcomes						
	Baseline Mean [SD]		8-Month Follow-Up Mean [SD]		Difference (95% CI)	P Value^a
Intervention group (n = 317)						
Systolic blood pressure mm Hg	145.5	[17.0]	134.2	[13.8]	11.3 (9.6-12.9)	<0.001
Diastolic blood pressure mm Hg	83.3	[10.4]	79.0	[8.6]	4.3 (3.2-5.4)	<0.001
Total cholesterol mg per dL	197.8	[38.5]	180.3	[26.6]	17.5 (14.2-21.0)	<0.001
Control group (n = 323)						
Systolic blood pressure mm Hg	143.0	[17.6]	138.2	[15.7]	4.8 (3.1-6.5)	<0.001
Diastolic blood pressure mm Hg	82.0	[11.3]	80.1	[9.8]	1.9 (0.8-3.0)	<0.001
Total cholesterol mg per dL	202.7	[40.8]	189.8	[30.9]	12.9 (9.6-16.1)	<0.001
Between-group comparison at 8-month follow-up						
	Intervention Group Mean [SD]		Control Group Mean [SD]		Difference (95% CI)	P Value^a
Systolic blood pressure mm Hg	134.2	[13.9]	138.2	[15.7]	4.0 (1.7-6.3)	0.001
Diastolic blood pressure mm Hg	79.0	[8.6]	80.1	[9.8]	1.1 (0.4-2.5)	0.158
Total cholesterol mg per dL	180.3	[26.6]	189.8	[30.9]	9.5 (5.1-14.0)	<0.001

^aMcNemar test to compare proportions or Student's t-test for paired samples to compare means for within-group changes from baseline to 8-month follow-up. Pearson chi-square test to compare proportions or the Student's t-test for independent samples to assess differences in means between the control and intervention groups.

^bBlood pressure lower than 140/90 mm Hg for patients with uncomplicated hypertension and lower than 130/80 mm Hg for those with diabetes, chronic kidney disease, or history of myocardial infarction or stroke.

^cCholesterol total lower than 200 mg per dL for patients without cardiovascular disease and lower than 175 mg per dL for patients with cardiovascular disease.

CI = confidence interval; mg per dL = milligrams per deciliter; mm Hg = millimeters mercury; SD = standard deviation.

The control group also showed statistically significant increases in the percentages of patients who achieved therapeutic goals in SBP (7.2 percentage points, 95% CI = 1.6-12.9); BP (7.2 percentage points, 95% CI = 1.7-12.8); and BP/TC (7.6 percentage points, 95% CI = 3.1-12.0; Table 2). However, the increases in the percentage of patients who achieved DBP therapeutic goals (2.8 percentage points, 95% CI = −2.5-8.1) and TC therapeutic goals (5.0 percentage points, 95% CI = −0.2-12.0) were not statistically significant.

At the 8-month follow-up, there were statistically significant

differences in favor of the intervention group compared with the control group in the percentage of patients at therapeutic goals for SBP (9.0 percentage points, OR = 1.43, 95% CI = 1.06-1.94); BP (9.5 percentage points, OR = 1.47, 95% CI = 1.08-1.99); TC (12.3 percentage points, OR = 1.64, 95% CI = 1.21-2.23); and BP/TC (15.3 percentage points, OR = 2.12, 95% CI = 1.50-2.98; Table 2, ORs not shown in table). However, the between-group difference in the percentage of patients at DBP therapeutic goal was not statistically significant (1.2 percentage points, OR = 1.06, 95% CI = 0.76-1.49).

Secondary Outcomes

The baseline mean [SD] values of SBP, DBP, and TC in the intervention group were 145.5 [17.0] mm Hg, 83.3 [10.4] mm Hg, and 197.8 [38.5] mg per dL, respectively. These values were not significantly different from those of the control group, where baseline values of SBP, DBP, and TC were 143.0 [17.6] mm Hg, 82.0 [11.3] mm Hg, and 202.7 [40.8] mg per dL, respectively (Table 2).

After 8 months of follow-up, patients in the intervention group showed statistically significant mean [SD] decreases in SBP (−11.3 [14.8] mm Hg), DBP (−4.3 [9.8] mm Hg), and TC (−17.5 [30.9] mg per dL; Table 2 [SDs not shown in table]). Similarly, patients in the control group showed statistically significant decreases in the mean values of SBP (−4.8 [15.4] mm Hg), DBP (−1.9 [10.0] mm Hg), and TC (−12.9 [29.8] mg per dL). At the 8-month follow-up, there were statistically significant differences in mean [SD]—SBP (−4.0 [14.8] mm Hg) and TC (−9.5 [28.9] mg per dL)—in favor of the intervention group compared with the control group. The difference in mean DBP was not statistically significant (−1.1 [9.2] mm Hg).

Discussion

The EMDADER-CV study was a large (n=714 patients) RCT designed to assess the effect of the Dader Method for pharmaceutical care on achieving therapeutic goals for BP, TC, and BP/TC in community pharmacy patients with CVD and/or high or intermediate CV risk. Compared with usual care (routine dispensing and oral counseling) plus written educational material about CV risk factors and CVD, pharmaceutical care produced significant improvements in the percentage of patients who achieved therapeutic goals for BP, TC, and BP/TC. It is important to point out that the Dader Method incorporated a combination of strategies that are considered key to achieving better control of CV risk, such as patient education on CV drugs, completion of a drug therapy profile and/or drug history, assessment of drug compliance, patient counseling about lifestyle modification, pharmacist-performed interventions not related to changes in drug therapy, and pharmacist-delivered treatment recommendations to physicians.^{27,31}

To our knowledge, there is a lack of large randomized trials designed to evaluate the effect of community pharmacist interventions in the achievement of therapeutic goals for BP, TC, and BP/TC in patients with CVD or CV risk factors, which was the main objective of the EMDADER-CV study. However, some studies, including some systematic reviews and meta-analysis,^{31–35} have demonstrated that pharmacist intervention could improve the outcomes of drug therapy among patients with hypertension^{17,18,21} and dyslipidemia.⁹

A prospective RCT by Morgado et al. (2011), designed to evaluate if a pharmaceutical care program (quarterly follow-up by a hospital pharmacist during a 9-month period) could improve BP control, evaluated a total of 197 patients, who were

randomly allocated to the intervention (n=98) or the control groups (n=99).¹⁷ At the end of the study, 63.3% (62/98) of the patients receiving pharmaceutical care compared with 43.4% (40/99) of control group patients reached the BP target ($P=0.005$); the intervention group also reached lower mean SBP (−6.8 mm Hg, $P=0.006$) and DBP (−2.9 mm Hg, $P=0.020$) than the control group (means of 134.2/82.5 mm Hg vs. 141.0/85.4 mm Hg, respectively). These values are higher than those in the EMDADER-CV study, in which 52.5% of patients in the intervention group achieved BP goals compared with 43.0% in the control group ($P=0.011$), and intervention group patients had a lower SBP (−4.0 mm Hg, $P=0.001$) but not a lower DBP (−1.1 mm Hg, $P=0.158$) than the control group (134.2/79.0 mm Hg vs. 138.2/80.1 mm Hg, respectively). Although these differences may be attributed to the *type of practitioners* (community pharmacists vs. hospital clinical pharmacists) and *setting* (community pharmacy vs. secondary care outpatient clinic in the university hospital), a more likely explanation is that, in the study by Morgado et al., only patients in the intervention group were provided with written educational material about hypertension and possible complications, as well as information on healthy lifestyle practices.¹⁷ By contrast, in the EMDADER-CV study, both the control and the intervention groups were provided with written educational material about CV risk factors and CVD and were focused on goals and the importance of adherence to pharmacologic and nonpharmacologic interventions to achieve therapeutic goals.³⁰ We consider that to be the reason why the EMDADER-CV control group showed statistically significant increases in the percentage of patients who achieved therapeutic goals for BP and decreases in the mean values of SBP and DBP.

In a 12-month, prospective, single-blind RCT by Hunt et al. (2008), designed to evaluate the effectiveness of collaborative management of hypertension by primary care-pharmacist teams in community-based clinics, patients with uncontrolled hypertension were randomized to a physician-pharmacist collaborative model of care intervention (n=230) or to usual primary care (n=233).¹⁸ According to an intention-to-treat analysis in which chart review was used to obtain the last recorded blood pressure values if the patient missed the last clinic visit, this study reported that 54% (125/230) of the patients in the intervention group achieved BP goals, compared with 42% (97/233) of the patients in the control group ($P=0.005$). Patients in the intervention group also had lower SBP (−6 mm Hg, $P=0.002$) and DBP (−3 mm Hg, $P=0.003$) compared with the control group (142/77 mm Hg vs. 148/80 mm Hg). The percentage of patients who achieved the targeted BP in the study by Hunt et al. was close to the value in the EMDADER-CV study; however, in the latter study, the intervention and the control groups were not significantly different in terms of DBP. The intervention in the physician-pharmacist collaborative model of care was similar to the intervention in the EMDADER-CV (the pharmacists reviewed subjects'

medications and lifestyle habits, assessed vital signs, screened for adverse drug reactions, identified barriers to treatment adherence, provided education, and scheduled follow-up appointments as deemed necessary). However, in the EMDADER-CV, pharmacists did not make direct alterations to drug regimens to titrate dosages, add a new agent, or switch a medication; instead, they sent written recommendations to the physician. By contrast, the physician-pharmacist collaborative arrangement studied by Hunt et al. permitted pharmacists to make drug regimen changes directly.¹⁸

A systematic review by Carter et al. (2009), including quasi-randomized trials, controlled before-after studies, interrupted time-series studies, patient-randomized trials, and cluster-randomized trials and designed to assess the effectiveness of interventions for BP involving nurses or pharmacists, showed that pharmacists' interventions were associated with a significant reduction in the mean value of SBP when the pharmacist made treatment recommendations to the physician (-9.30 mm Hg) or when the pharmacist performed the intervention (-8.44 mm Hg).³¹ In a nonparametric analysis of 13 studies involving community pharmacists, the mean [SD] reductions in SBP and DBP were -9.31 [5.00] and -4.59 [4.64] mm Hg.³¹ These values are close to the results of the EMDADER-CV study, in which the intervention group experienced mean [SD] reductions in SBP and DBP of -11.3 [14.8] and -4.3 [9.8] mm Hg, respectively. Additionally, based on the results of 5 studies conducted in community pharmacies, Carter et al. estimated an OR for controlled BP of 2.89 (95% CI=1.83-4.55) for community pharmacists, a value higher than the OR for controlled BP in our study (1.47, 95% CI=1.08-1.99). However, Carter et al. clarified that if 2 studies were excluded (1 because of an extremely high OR=29.7 and the other because the pharmacist worked closely with 2 physicians and reviewed medical records of study participants in the physicians' office, OR=4.29), and 1 additional study was included (classified as a nursing intervention for the OR calculations, but the intervention involved both a nurse and a community pharmacist, OR=1.79), the estimated OR for controlled BP would have been 1.8,³¹ a value that would be closer to the OR obtained in the EMDADER-CV study (1.47).

In the EMDADER-CV study, the difference in mean SBP in favor of the intervention group compared with the control group was lower than that of -6.9 [12.0] mm Hg ($P=0.047$) found in the systematic review by Machado et al. (2007)³² but was close to the value of -4.9 [0.9] mm Hg ($P<0.001$) obtained in the systematic review by Morgado et al. (2011).³³ By contrast, neither the EMDADER-CV study nor the review by Machado et al.³² found significant differences between the intervention group and the control group in DBP, whereas the difference of -2.6 [0.9] mm Hg ($P<0.001$) was significant in the meta-analysis by Morgado et al.³³ In the meta-analyses by Machado et al.³² and by Morgado et al.,³³ the control group patients did

not experience a significant reduction in the mean value of DBP (-2.4 mm Hg [95% CI= -6.1 to 1.3] and -4.9 [3.0] mm Hg [$P=0.103$], respectively), while in the present study, the control group patients experienced a significant decrease in the mean value of DBP. In addition to differences in terms of the type of studies (meta-analysis vs. only 1 RCT), it is important to point out that both the control and the intervention groups in the EMDADER-CV study were provided with written educational material about CV risk factors.³⁰

In a recent systematic review and meta-analysis of 30 RCTs (11,765 patients) by Santschi et al. (2011), conducted to determine the impact of pharmacist care on the management of CV risk factors among outpatients, pharmacist care was associated with significant reductions in mean values of SBP and DBP (-8.1 mm Hg, 95% CI= -10.2 to -5.9 and -3.8 mm Hg, 95% CI= -5.3 to -2.3 , respectively).³⁴ These results are considerably higher than the reductions obtained in both SBP and DBP in the EMDADER-CV study and in the systematic reviews conducted by Machado et al.³² and Morgado et al.³³ Differences obtained in the systematic reviews by Machado et al.,³² Morgado et al.,³³ and Santschi et al.³⁴ might be showing a tendency to an overall increment in treatment adherence in the most recent studies, which have tended to favor pharmacist care compared with usual care and included more interventions involving physician-pharmacist collaboration³³ and a substantial heterogeneity.³⁴ Despite these quantitative differences, all these studies, including EMDADER-CV, found that community pharmacy interventions improve the outcomes of drug therapy among patients with hypertension.^{17,18,31-34}

In the Study of Cardiovascular Risk Intervention by Pharmacists (SCRIP), an RCT of a community pharmacist intervention in primary care of patients with dyslipidemia, Tsuyuki et al. (2002) assigned 675 patients with CVD or diabetes with other CV risk factors to a pharmacist intervention including education and a brochure about CV risk factors; point-of-care cholesterol measurement; referral to their physicians; regular follow-up for 16 weeks; and risk factor information provided by the pharmacist to the physician, or to usual care including the brochure, general advice, and minimal follow-up.⁹ The primary endpoint (performance of a fasting cholesterol panel by the physician or addition or increase in dose of cholesterol-lowering medication) was reached in 196 patients (57%) in the intervention group, compared with 102 (31%) in the usual care group. Although the primary end point in the SCRIP was different than the goals established for TC in the EMDADER-CV study, the percentage of patients in the intervention group who reached the primary end point in the SCRIP was similar to the percentage of patients who reached therapeutic goals for TC in our study.⁹

The reduction in TC obtained in the intervention group of the EMDADER-CV study is considerably lower than the

reduction found in the pharmacist intervention group in the systematic review and meta-analysis in hyperlipidemia management by Machado et al. (2008),³⁵ a mean [SD] of -34.2 [10.3] mg per dL ($P < 0.001$), which included a total of 2,084 patients (685 in the pharmacist intervention group and 1,399 in the control group). Although the significant reduction in TC obtained in the control group of the EMDADER-CV study is similar to the nonsignificant reduction in the control group of the review by Machado et al.,³⁵ a mean [SD] of -13.7 (10.3) mg per dL ($P = 0.186$), the mean difference between intervention and control groups in the present study is considerably smaller than the one reported by Machado et al.,³⁵ -22.0 (10.4) mg per dL ($P = 0.034$). In the systematic review by Santschi et al.³⁴ based on 9 studies (1,121 patients), pharmacist care was associated with a significant reduction in TC, a mean of -17.4 mg/dL (95% CI = -25.5 to -9.2); those results are close to the reduction found by Machado et al.³⁵ but higher than the value found in the present study. Despite some quantitative differences, evidence suggests that community pharmacist intervention is beneficial for patients with CVD or CV risk factors.^{9,34,35}

Limitations

First, the results might have been subjected to potential bias. Because there was no “placebo” treatment, the study patients were not blinded to treatment assignment, and both groups (intervention and control) were seen by the same community pharmacist, who was responsible for collecting BP and TC measurements. However, potential bias was reduced by pharmacist use of a standardized process for collecting BP and TC measurements in both the intervention and control groups.

Second, control group patients received telephone calls from the pharmacist, who encouraged their attendance at the appointments. Control group patients were also provided with written and oral education about CV risk factors and CVD and were counseled on goals and the importance of adherence to pharmacological and nonpharmacological interventions to achieve therapeutic goals, as requested by the Human Research Ethical Committee in its approval of this research project. The committee also requested that pharmacists provide patients with the results of BP and TC assessments. In addition, all patients were questioned about knowledge of CVD, and 41 were questioned about treatment adherence. Therefore, control group patients received some intervention, and 10 patients allocated to the control were excluded from the study because pharmacists determined that the patients needed intervention. The effect of written educational material and the extra attention received by the patients in the usual care group might have produced positive outcomes and reduced the magnitude of differences compared with the intervention group.

The third potential limitation was the final sample size (714 patients), which was smaller than expected (752 patients). The

size was estimated assuming that in the control group only 30% of the patients would achieve therapeutic goals for BP and TC. This assumption was made based on individual values of BP and TC control reported in previous work.⁶⁻⁹ However, during the progress of the EMDADER-CV study, some published studies with detailed information from Spain showed that the percentage of patients who achieved therapeutic goals for both BP and TC was less than 20%.³⁶ Thus, if we assumed that in the intervention group this percentage would be improved from 20% to 30%, with a significance level of 0.05 (2-tailed) and a power of 80%, we would estimate a required sample of 626 patients (313 in each group). As a consequence, a final sample size of 714 patients (356 in the intervention group and 358 in the control group) would be considered as appropriate for the hypothesis and the primary outcomes of the EMDADER-CV study.

Finally, the EMDADER-CV study was performed by carefully selected community pharmacists, who were trained and specialized in pharmacotherapy follow-up and in using the Dader Method, which limits the degree to which the present study findings can be generalized to other settings.

Conclusions

Compared with usual care plus written education, pharmaceutical care focused on patient evaluation and follow-up in collaboration with physicians improved the achievement of BP, TC, and BP/TC treatment goals in patients with CVD and/or high or intermediate CV risk in a community pharmacy setting in Spain. The EMDADER-CV study demonstrates the added value of pharmaceutical care by community pharmacists in outpatients with CVD and/or high or intermediate CV risk.

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Effectiveness of Dader Method for Pharmaceutical Care on Control of Blood Pressure and Total Cholesterol in Outpatients with Cardiovascular Disease or Cardiovascular Risk: EMDADER-CV Randomized Controlled Trial

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